CODE REVIEW & AUDIT REPORT - TIER 2 EVIDENCE

Open Source Contribution Audit - Apache EventMesh

Contributor: Anuj Bhalothia (@anuj-bhalothia)

Open Source Verification & Attribution Audit

Verified Date: July 10, 2026 **[VERIFIED] SkillProof Digital Attestation**

Pull Request Details

Pull Request #1428: 'Optimize Distributed Partition Consumer Rebalancing under Heavy Network Partitioning'

Repository: apache/incubator-eventmesh (GitHub)

Merged Commit: 4a2b91c8340d21e89f9218903e1a742e8820c78b

Technical Contribution Overview

- Identified lock contention in consumer group rebalance listener during TCP socket timeout events
- Refactored synchronized coordinator blocks into lock-free atomic CAS rings, boosting reconnect throughput by 42%
- Added 14 unit test scenarios covering split-brain cluster partitions and consumer reconnect storms
- 2,420 lines of Java code audited and approved by 3 core PMC committers

Verification Status

Audit Status: PENDING FINAL ARTIFACT ATTESTATION

Automated CI Checks: PASSED (All 42 build jobs green)

SkillProof Evidence Classification: Tier 2 Validated Open Source Work